

WHEN SITTING STILL LEADS TO A STANDSTILL

SEDENTARY TIME PREDICTS A HIGHER RISK OF TRANSITION FROM HEALTHY TO BURNOUT-RELATED COPING PATTERNS

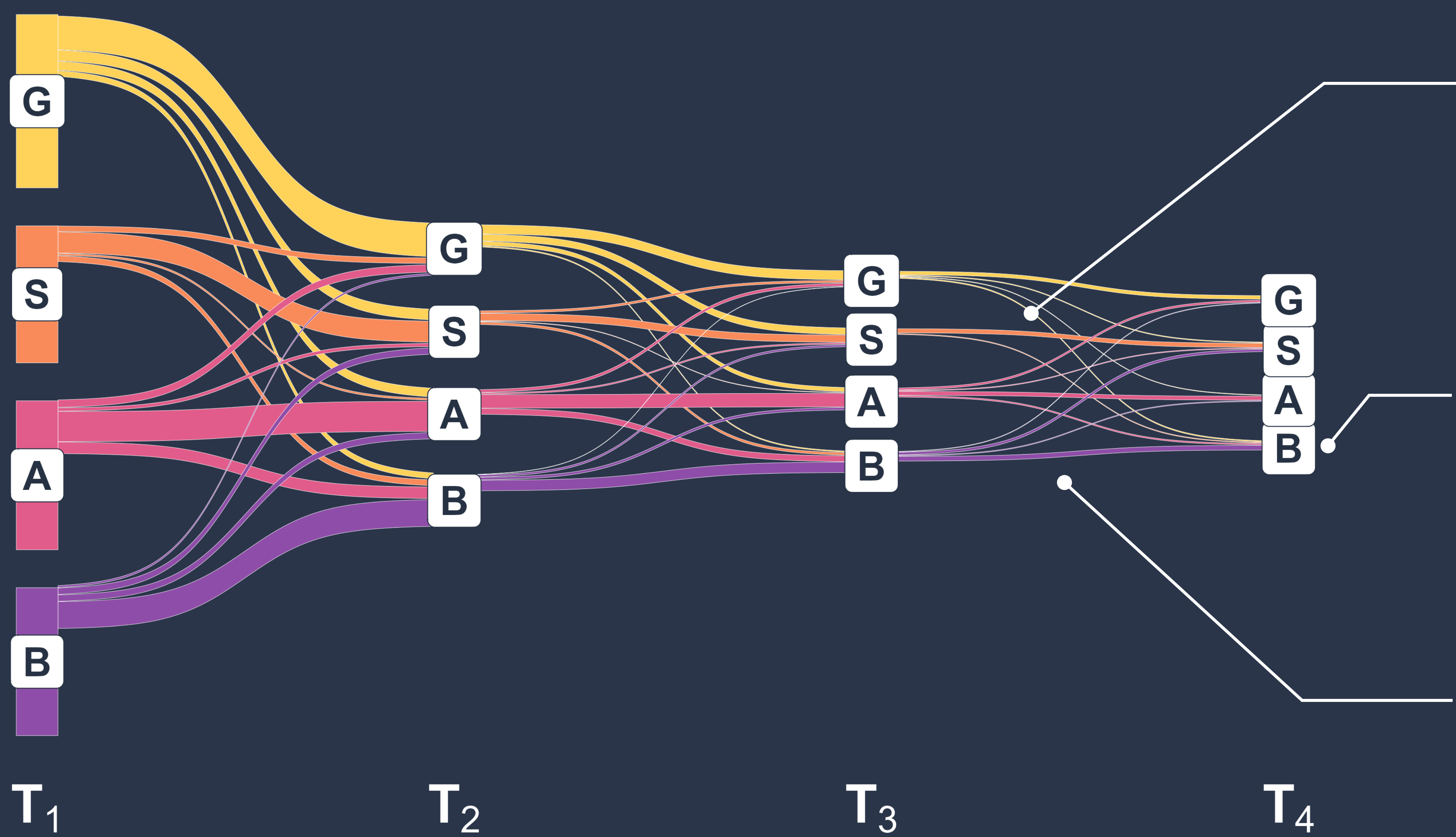


Figure 1. Proportion of students in each coping pattern at T₁–T₄ (person-centered time points aligned with each participant's first participation between 2022 – 2025), indicated by node widths, and their proportional transitions across consecutive years, indicated by flow widths.

Background

University students face considerable academic and psychosocial stressors, placing them at elevated risk for mental health challenges compared to non-student peers (Sheldon et al., 2021). Physical activity (PA) has been shown to buffer stress and promote well-being in students, whereas sedentary time (ST) is linked to adverse health outcomes (Wunsch et al., 2021). However, little is known about how PA and ST relate to patterns of academic behavior and experience reflecting students' coping with academic demands.

Research Question

How are PA and ST associated with transitions between academic coping behavior and experience patterns in students over time?

Results

1 Sedentary Time

Higher sedentary time was associated with transitions from healthy (G) to burnout-related coping (B) within one academic year ($HR = 2.30$, 95% CI: 1.15 – 4.59).

2 Transition Probability

The probability of B (Burnout) being the next pattern after G (Good Health) was 3% (95% CI: 1% – 22%) for low (–1 SD) and 16% (95% CI: 6% – 38%) for high (+1 SD) sedentary time.

3 Physical Activity

Physical activity was not significantly associated with transition hazard ratios between coping patterns ($HRs = 0.77 – 1.29$, all 95% CIs included 1).

Method

Outcome:

Coping Behavior and Experience Patterns (WCEP, Schaarschmidt & Fischer, 2008)



Discussion

Long periods of sitting indicate working conditions characterized by high cognitive demands, few breaks, and monotonous tasks - factors that contribute to stress and limited recovery. Reducing structurally mandated sitting could support healthier coping patterns in academic settings.

Predictors

PA: Moderate to vigorous intensity PA in metabolic equivalent of task minutes per week (IPAQ, Craig et al., 2003)

ST: Daily sitting time in minutes (IPAQ, Craig et al., 2003)

Data Analysis

Continuous-time multi-state Markov models were applied to estimate the effects of PA and ST on transition probabilities between patterns over time.

Limitations

- If dropout is related to coping state, transition estimates may be biased.
- Few events for some transitions limit precision.
- With one year between survey waves, transitions between intervals are unobserved.
- Self-reported data and observational design.

Design

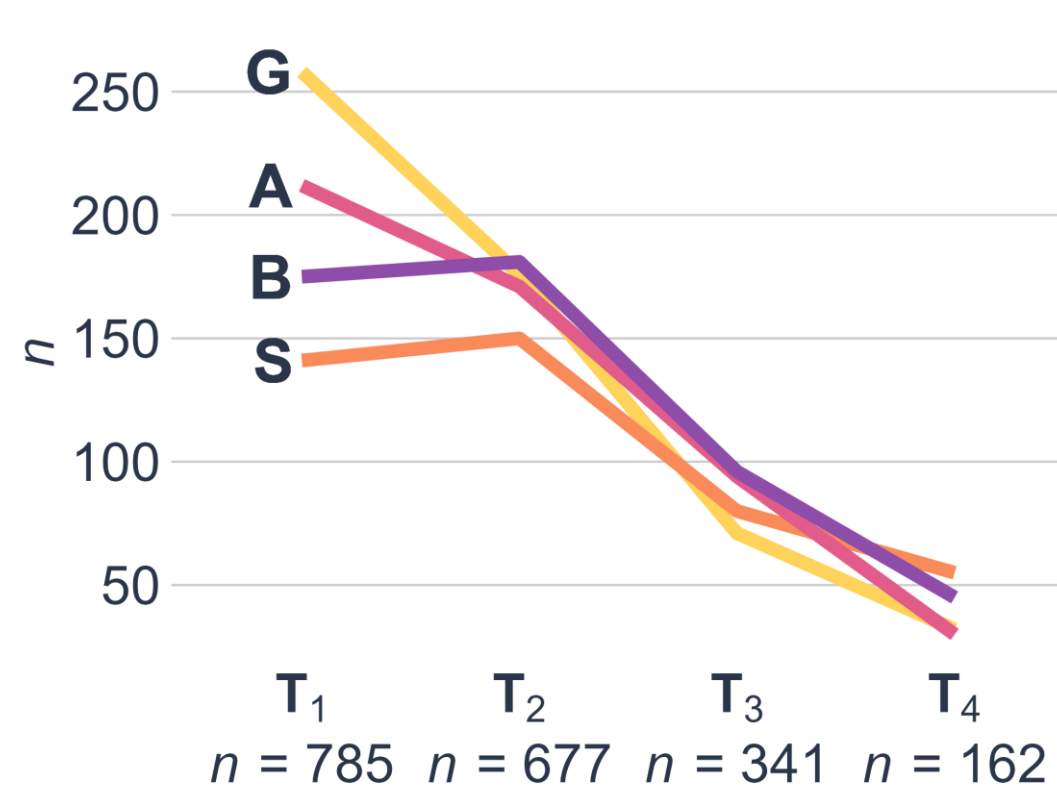


Figure 3. n per pattern over time (2022 – 2025). Longitudinal survey.

Learn More



Poster, Contact, References, & Supplement